Assignment 1

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Part 1: Benchmarking

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OUTPUT    PORTS    TERMINAL    PROBLEMS

(.venv) PS C:\Users\qhung\Videos\CSIS4260 - Special Topic in Data Analytics\Assignments\HughTran_Assignment1>
          date   open   high    low  close    volume name
0  2013-02-08  15.07  15.12  14.63  14.75   8407500  AAL
1  2013-02-11  14.89  15.01  14.26  14.46   8882000  AAL
2  2013-02-12  14.45  14.51  14.10  14.27   8126000  AAL
3  2013-02-13  14.30  14.94  14.25  14.66  10259500  AAL
4  2013-02-14  14.94  14.96  13.16  13.99  31879900  AAL

DataFrame info():
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 619040 entries, 0 to 619039
Data columns (total 7 columns):
 #   Column  Non-Null Count   Dtype
---  ------  --------------   -----
 0   date    619040 non-null  object
 1   open    619029 non-null  float64
 2   high    619032 non-null  float64
 3   low     619032 non-null  float64
 4   close   619040 non-null  float64
 5   volume  619040 non-null  int64
 6   name    619040 non-null  object
dtypes: float64(4), int64(1), object(2)
memory usage: 33.1+ MB

==============================
A. CSV: Measure size + read time (1x, 10x, 100x)
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Convert CSV -> Parquet (1x) and measure conversion time
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B. Parquet: Measure size + read time (1x, 10x, 100x)
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DONE! Results saved
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Results file: outputs/part1_benchmarking.csv

Preview results:
    format scale compression                              path  write_time_sec  size_bytes  read_time_sec
0      csv    1x        none             data/all_stocks_5yr.csv        0.000000    30098860       0.312105
1      csv   10x        none                 temp_folder\csv_10x        0.271747   300988600       3.187757
2      csv  100x        none                temp_folder\csv_100x        4.132972  3009886000      32.189963
3  parquet    1x      snappy  temp_folder\data_1x_snappy.parquet        0.670103    10521335       0.169099
4  parquet   10x      snappy             temp_folder\parquet_10x        0.049047   105213350       0.624618
5  parquet  100x      snappy            temp_folder\parquet_100x        1.577916  1052133500       7.028601

x10/x100 files deleted.
```

Based on the benchmarking results, the Parquet approach demonstrated better performance in terms of reading time, writing time, and file size. For the current dataset size, the differences between CSV and Parquet are not very significant. Therefore, using

CSV would still be acceptable for now. However, from a long-term perspective, as the dataset scales up, Parquet would be a more suitable choice due to its superior storage efficiency and performance advantages.

Part 2: Model training

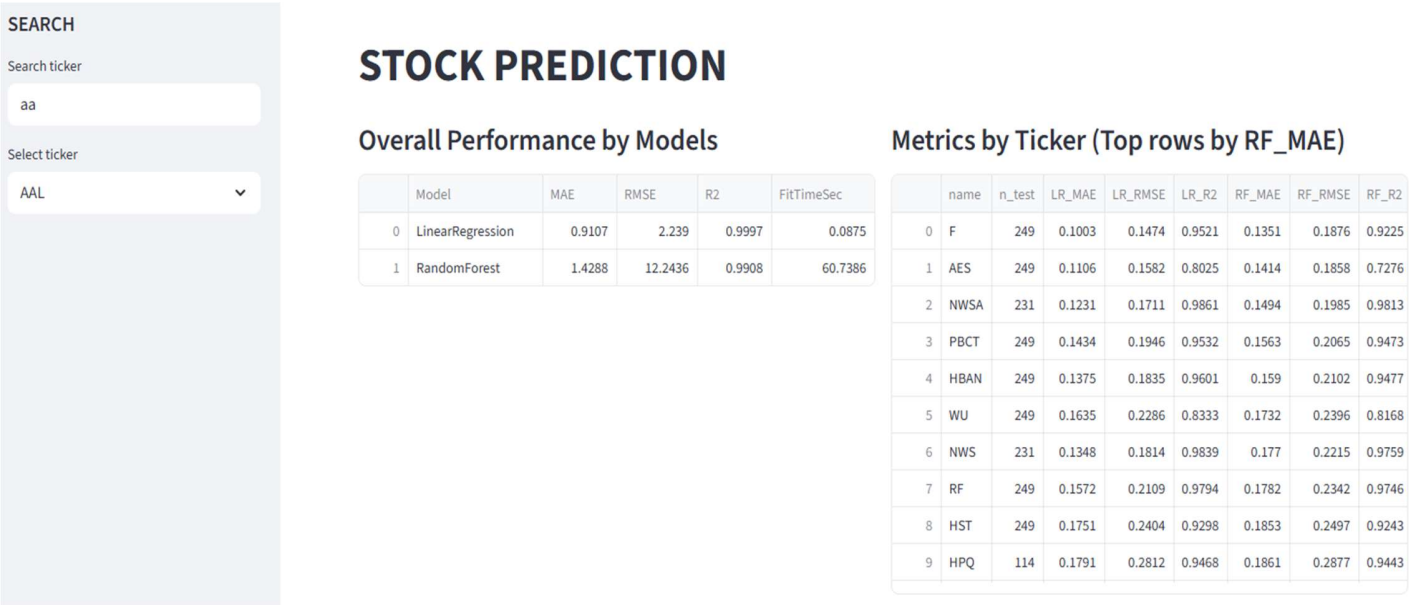
Linear Regression was selected as a simple model because it is easy to understand and very fast to run. It helps show the general relationship between the technical indicators and the next-day closing price. Random Forest was selected because it is a more complex model that can learn more complicated patterns in the data. By comparing these two models, we can see whether stock prices follow a simple linear pattern or a more complex pattern.

The results show that Linear Regression performs better than Random Forest in this case. It has lower MAE and RMSE values, which means the prediction errors are smaller. It also has a higher R^2 value, which means it explains more of the changes in the data. In addition, Linear Regression takes much less time to train compared to Random Forest.

This result makes sense because the next-day closing price is usually very close to today's closing price. Since the relationship between these prices is mostly linear, a simple linear model can work very well. The Random Forest model is more complex, but in this situation, the extra complexity does not improve the performance.

Part 3: Dashboard

Streamlit was chosen because it is simple and easy to use. It allows us to build an interactive dashboard using only Python, without needing extra web programming skills like HTML or JavaScript. It provides built-in tools such as search boxes, selection menus, and charts..



	Model	MAE	RMSE	R2	FitTimeSec
0	LinearRegression	0.9107	2.239	0.9997	0.0875
1	RandomForest	1.4288	12.2436	0.9908	60.7386

	name	n_test	LR_MAE	LR_RMSE	LR_R2	RF_MAE	RF_RMSE	RF_R2
0	F	249	0.1003	0.1474	0.9521	0.1351	0.1876	0.9225
1	AES	249	0.1106	0.1582	0.8025	0.1414	0.1858	0.7276
2	NWSA	231	0.1231	0.1711	0.9861	0.1494	0.1985	0.9813
3	PBCT	249	0.1434	0.1946	0.9532	0.1563	0.2065	0.9473
4	HBAN	249	0.1375	0.1835	0.9601	0.159	0.2102	0.9477
5	WU	249	0.1635	0.2286	0.8333	0.1732	0.2396	0.8168
6	NWS	231	0.1348	0.1814	0.9839	0.177	0.2215	0.9759
7	RF	249	0.1572	0.2109	0.9794	0.1782	0.2342	0.9746
8	HST	249	0.1751	0.2404	0.9298	0.1853	0.2497	0.9243
9	HPQ	114	0.1791	0.2812	0.9468	0.1861	0.2877	0.9443

The selected ticker: AAL

Price Predictions

Technical Indicators

EMA

MACD
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Mar 05
Apr 02
Apr 30
May 28
Jun 25
Jul 23
Aug 20
Sep 17
Oct 15
Nov 12
Dec 10
Jan 07
date Dec 21, 2017
value 0.702112774375
color MACD_Signal
MACD
MACD_Hist
MACD_Signal

RSI
100
90
80
70
60
50
40
30
20
10
0
Feb 12
Feb 26
Mar 12
Mar 26
Apr 09
Apr 23
May 07
May 21
Jun 04
Jun 18
Jul 02
Jul 16
Jul 30
Aug 13
Aug 27
Sep 10
Sep 24
Oct 08
Oct 22
Nov 05
Nov 19
Dec 03
Dec 17
Dec 31
Jan 14
Jan 28
date Nov 22, 2017
RSI_14 61.7753623188